10TH CLASS COMPUTER SCIENCE — NEW SYLLABUS (PECTAA)

Chapter 3 — Introduction to Python Programming

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Multiple Choice Questions

1. Which of the following steps is NOT part of the basic programming process?

(a) Write Code (b) Compile/Interpret
(c) Execute (d) Ignore Errors

Answer: (d) — The basic programming process is: write the code, compile or interpret it, execute it and then remove the errors. Ignoring the errors is not a step of this process, because a program with errors cannot give the correct output.

2. What should you do when installing Python to run it from the command line more easily?

(a) Uncheck “Add Python to PATH” (b) Choose a different IDE
(c) Check “Add Python to PATH” (d) Install only the IDE

Answer: (c) — During the installation, the check box “Add Python to PATH” must be checked. In this way the Python interpreter is added to the system path and the command python can be run from the command line from any folder.

3. What will be the output of: print(10 // 3)?

(a) 3.333 (b) 3
(c) 1 (d) 4

Answer: (b) — The // operator is the floor division operator, which divides the numbers and gives only the whole-number part of the result. So 10 // 3 gives 3, while 10 / 3 gives 3.333 and 10 % 3 gives the remainder 1.

4. What is data type of variable x in x = “123”?

(a) Integer (b) Float
(c) String (d) Boolean

Answer: (c) — The value 123 is written inside quotation marks, and in Python anything written inside quotes is a string (str). It looks like a number, but it is text, so it cannot be used in arithmetic until it is converted with int().

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5. Which operator has the highest precedence in Python?

(a) Addition (+) (b) Multiplication (*)

(c) Exponentiation (**) (d) Parentheses ()

Answer: (d) — The operations written inside parentheses are performed first of all, so the parentheses have the highest precedence. After them come the exponentiation, then the multiplication, division and modulus, and at the end the addition and subtraction.

6. What is the result of 5 > 3 and 2 < 1?

(a) True (b) False

(c) Error (d) None

Answer: (b) — 5 > 3 is True, but 2 < 1 is False. The logical operator and gives True only when both conditions are True, so the whole expression becomes False.

7. What does the += operator do?

(a) Adds two numbers (b) Compares values

(c) Adds and assigns (d) Multiplies values

Answer: (c) — The += operator is an assignment operator. It adds the value on the right to the variable on the left and then stores the new value in the same variable, so a += 5 is the same as a = a + 5.

8. What is the output of the below code? age = 25; print("Age :", age)

(a) Age: 25 (b) 25

(c) Age (d) age

Answer: (a) — The print() function displays all its arguments in the same order. The text “Age :” is printed as it is, and after it the value of the variable age, that is 25, is printed, so the output is: Age: 25.

9. Which symbol is used for comments in Python?

(a) // (b) /*

(c) # (d) --

Answer: (c) — A single-line comment in Python starts with the hash symbol (#). The interpreter does not execute the text written after it. The symbols // and /* belong to other languages such as C and Java.

B. Short Answer Questions

1. Why are comments important in programming?

Comments are the lines of the program which are not executed by the Python interpreter; a single-line comment starts with the # symbol. They are important because they explain what the code is doing, so the programmer and other readers can understand the program easily. Comments also make the code easier to check and to change later, and they can be used to leave notes about the purpose of a variable or of a difficult step. In short, comments improve the readability of the program without affecting its output.

2. Discuss three basic data types in Python.

(i) Integer (int): it stores whole numbers, without any decimal point; for example **age = 17**.

(ii) Float: it stores decimal numbers, that is, numbers with a fractional part; for example **price = 19.99**.

(iii) String (str): it stores text, which is written inside single or double quotation marks; for example **name = “Ali”**.

A fourth common type is **Boolean (bool)**, which stores only **True** or **False**, for example **is_pass = True**. Python itself decides the data type of a variable from the value which is assigned to it.

3. Describe the difference between integer and float data types in Python.

Integer (int)	Float
It stores whole numbers only.	It stores numbers which have a decimal part.
It does not use a decimal point.	It always contains a decimal point.
Example: age = 17, marks = 480, count = -5	Example: price = 19.99, height = 1.65, temp = -2.5
It uses less memory and the calculation is exact for whole numbers.	It uses more memory, and some decimal values are stored in an approximate form.

4. What are logical operators and name all three?

Logical operators are the operators which are used to combine two or more conditions or expressions, and they return a Boolean value, that is True or False. The three logical operators in Python are:

(i) and: it gives True only when both conditions are True; for example 5 > 3 and 2 < 4 gives True.

(ii) or: it gives True when at least one condition is True; for example 5 > 3 or 2 > 4 gives True.

(iii) not: it reverses the result, so not True gives False and not False gives True.

5. How do you get user input in Python?

User input is taken with the **input()** function. This function shows a message on the screen and waits until the user types something and presses Enter. Whatever the user types is stored as a string in a variable. For example: **name = input(“Enter your name: ”)**. If a number is needed, the string is converted by using the **int()** function for whole numbers or the **float()** function for decimal numbers, for example **age = int(input(“Enter your age: ”))**.

6. What is the purpose of print() statement in Python?

The **print()** function is used to display information on the screen, that is, to give the output of the program to the user. It can take one or more arguments, which are printed in the same order, and it can also print text, numbers and the values of variables together. For example **print("Hello, Students!")** shows the text on the screen, while **print("Your age is:", age)** shows the message along with the value stored in the variable age.

7. What is operator precedence? Give an example.

Operator precedence means the order in which the operations of an expression are performed when the expression contains more than one operator. In Python, the parentheses () have the highest precedence, then the exponentiation **, then the multiplication *, division /, floor division // and modulus %, and at the end the addition + and subtraction −. For example, in **4 * 3 + 10 / 2 − 11 % 3** the multiplication, division and modulus are done first, so the expression becomes 12 + 5.0 − 2 and the answer is 15.0. If we write **(3 + 2) * 4**, the answer is 20, because the parentheses are solved first.

8. Write any three main rules for naming variables in Python.

(i) The name must begin with a letter (a–z, A–Z) or an underscore (_); it cannot begin with a digit.

(ii) After the first character, the name may contain letters, digits (0–9) or underscores, but no space or special symbol.

(iii) Variable names are case-sensitive, so age and Age are two different variables, and the reserved keywords of Python, such as if, else and for, cannot be used as variable names.

C. Long Answer Questions

1. What are the basic data types in Python? Explain integers, floats, strings, and Booleans with examples.

A data type tells us which kind of value a variable can store. Python itself decides the data type of a variable from the value which is given to it, and the four basic data types are the following.

Data type	What it stores	Example
Integer (int)	Whole numbers, positive, negative or zero, without a decimal point.	age = 17 marks = 480
Float	Numbers with a decimal part, also called floating-point numbers.	price = 19.99 height = 1.65
String (str)	Text, written inside single or double quotation marks. It may contain letters, digits and symbols.	name = "Ali" city = 'Lahore'
Boolean (bool)	Only two values: True or False. It is used in conditions and comparisons.	is_pass = True found = False

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Examples in a program:

Code

```
age = 17              # int
price = 19.99         # float
name = "Ali"          # str
is_pass = True        # bool

print(type(age))
print(type(price))
print(type(name))
print(type(is_pass))
```

Output:

```
&lt;class 'int'&gt;
&lt;class 'float'&gt;
&lt;class 'str'&gt;
&lt;class 'bool'&gt;
```

The **type()** function tells the data type of a value. If a number is written inside quotation marks, such as x = “123”, it becomes a string and not an integer, so it cannot be used in calculations until it is converted with int().

2. What are arithmetic operators in Python? Provide examples of each operator and explain when you would use them.

Arithmetic operators are the symbols which are used to perform the mathematical operations on numbers. The arithmetic operators of Python, with their examples, are given below.

Operator	Meaning	Example (a = 10, b = 3)	Result
+	Addition	a + b	13
−	Subtraction	a − b	7
*	Multiplication	a * b	30
/	Division (gives a float)	a / b	3.3333
//	Floor division (whole part only)	a // b	3
%	Modulus (remainder)	a % b	1
**	Exponentiation (power)	a ** b	1000

When they are used:

(i) + and − are used to add and subtract the values, for example to find the total marks or the remaining amount.

(ii) * and / are used to find the product and the average, for example the area of a room or the average marks of a student.

(iii) // is used when only the whole number is needed, for example how many full packets can be made from the total items.

(iv) % is used to find the remainder, for example to check whether a number is even or odd.

(v) ** is used to find the power of a number, for example the square or the cube of a value.

Program

```
a = 10
b = 3
print(a, "+", b, "=", a + b)     # 10 + 3 = 13
print(a, "/", b, "=", a / b)     # 10 / 3 = 3.3333333333333335
print(a, "//", b, "=", a // b)   # 10 // 3 = 3
print(a, "%", b, "=", a % b)     # 10 % 3 = 1
print(a, "**", b, "=", a ** b)   # 10 ** 3 = 1000
```

3. Explain the input() and print() functions with examples.

(a) input() function: the input() function is used to take data from the user. It displays a message on the screen, waits for the user to type something and press Enter, and then stores the typed text in a variable. Whatever the user enters is always taken as a string, so for numbers we use int() for whole numbers and float() for decimal numbers.

Program

```
name = input("Enter your name: ")
print("Hello, " + name + "!")

age = int(input("Enter your age: "))
print("Your age is:", age)
```

Output

```
Enter your name: Maryam
Hello, Maryam!
Enter your age: 16
Your age is: 16
```

(b) print() function: the print() function is used to display information on the screen. It can take one or more arguments, separated by commas, and it prints them in the same order. Text is written inside quotation marks, while a variable is written without quotes so that its value is printed.

Program

```
print("Hello, Students!")
age = 17
print("Age :", age)
print("Total =", 4 + 5)
```

Output

```
Hello, Students!
Age : 17
Total = 9
```

In this way input() takes the data into the program and print() shows the result to the user, so both functions together make the program interactive.

4. Explain how comparison and logical operators work together in conditional statements. Provide a practical example.

Comparison operators compare two values and return a Boolean result, that is True or False. They are >, <, >=, <=, == and !=. For example, if x = 10 and y = 5, then x > y gives True and x == y gives False.

Logical operators (and, or, not) join two or more such conditions into one condition. The **and** gives True only when both conditions are True, the **or** gives True when at least one of them is True, and the **not** reverses the result.

When these operators are used in a conditional statement such as **if**, the whole condition is checked and the block of code runs only when the final result is True, otherwise the **else** block runs. In this way the program takes a decision.

Program

```
age = int(input("Enter your age: "))
citizen = input("Do you have a CNIC? (yes/no): ")

if age &gt;= 18 and citizen == "yes":
    print("You can apply for a driving licence.")
else:
    print("You are not eligible for a driving licence.")
```

Output

```
Enter your age: 20
Do you have a CNIC? (yes/no): yes
You can apply for a driving licence.
```

Explanation: here two comparisons are made: age >= 18 and citizen == “yes”. The logical operator **and** joins them, so the message of the **if** block is printed only when the person is 18 years or older and also has a CNIC. If any one of the two conditions is False, the **else** block is printed. This is how the comparison and the logical operators work together in the conditional statements.

5. Write a program in Python that will take a name from user and display greeting message to the given name.

Program

```
name = input("Enter your name: ")
print("Assalam-o-Alaikum, " + name + "!")
print("Welcome to Python programming.")
```

Output

```
Enter your name: Ayesha
Assalam-o-Alaikum, Ayesha!
Welcome to Python programming.
```

The input() function takes the name from the user and stores it in the variable name. The print() function then joins the greeting with the value of that variable with the help of the + operator and shows the message on the screen.

6. Write a program in Python that takes two numbers from user and display their sum.

Program

```
num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

total = num1 + num2
print("The sum of", num1, "and", num2, "is", total)
```

Output

```
Enter first number: 15
Enter second number: 25
The sum of 15 and 25 is 40
```

Both numbers are taken with input() and converted into integers with int(), because the input() function returns a string. If the numbers were not converted, the + operator would join them as text and give 1525 instead of the sum 40.

7. Write a program in Python that takes five numbers from user and display their average.

Program

```
n1 = float(input("Enter number 1: "))
n2 = float(input("Enter number 2: "))
n3 = float(input("Enter number 3: "))
n4 = float(input("Enter number 4: "))
n5 = float(input("Enter number 5: "))

total = n1 + n2 + n3 + n4 + n5
average = total / 5

print("Total =", total)
print("Average =", average)
```

Output

```
Enter number 1: 10
Enter number 2: 20
Enter number 3: 30
Enter number 4: 40
Enter number 5: 50
Total = 150.0
Average = 30.0
```

The float() function is used because the numbers may contain decimals, and the average is found by dividing the total by 5.

8. Solve the following expressions.

(a) 8 + 3 * 4 − 2 ** 2

= 8 + 3 * 4 − 4 (2 ** 2 = 4, power first)

= 8 + 12 − 4 (3 * 4 = 12, multiplication before addition)

= **16**

(b) 20 % 7 * 3 + 10 // 3

= 6 * 3 + 10 // 3 (20 % 7 = 6, modulus first)

= 18 + 3 (6 * 3 = 18 and 10 // 3 = 3)

= **21**

(c) 2 * 3 ** 2 % 4 + 10 − 6 / 2

= 2 * 9 % 4 + 10 − 3.0 (3 ** 2 = 9 and 6 / 2 = 3.0)

= 18 % 4 + 10 − 3.0 (2 * 9 = 18)

= 2 + 10 − 3.0 (18 % 4 = 2)

= **9.0**

In every part the order of precedence is followed: parentheses first, then exponentiation, then multiplication, division, floor division and modulus from left to right, and at the end addition and subtraction.

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